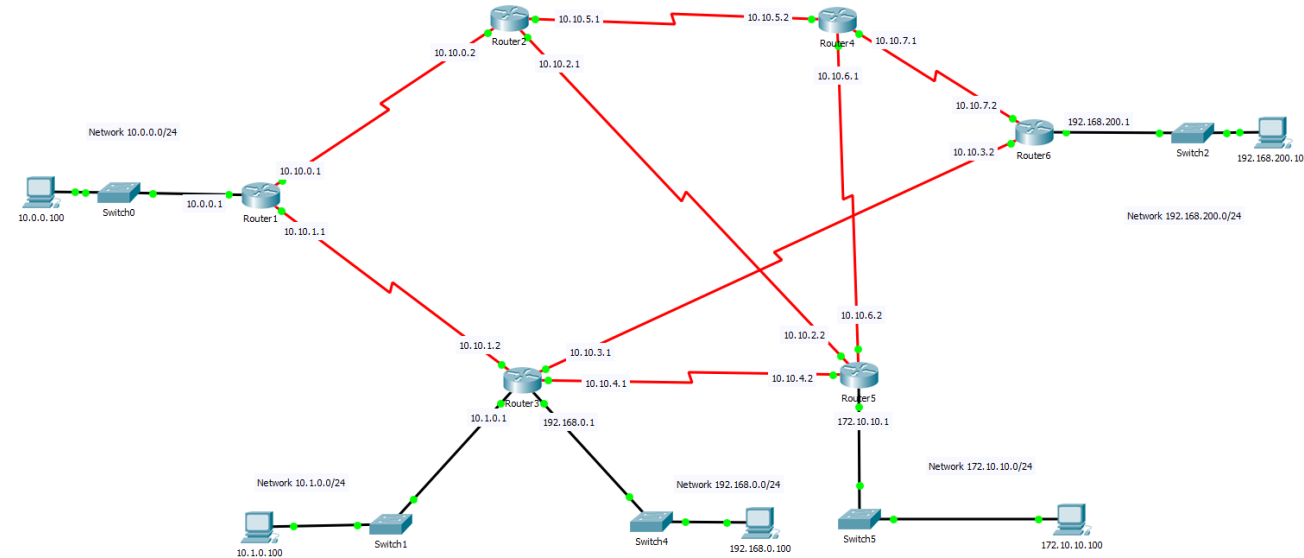


Network Fundamentals

IPv6

Internet protocol

- Addressing end devices
- Routing packets through the network
- Keep information about the network using routing tables
- IPv4 and IPv6 have the same responsibilities



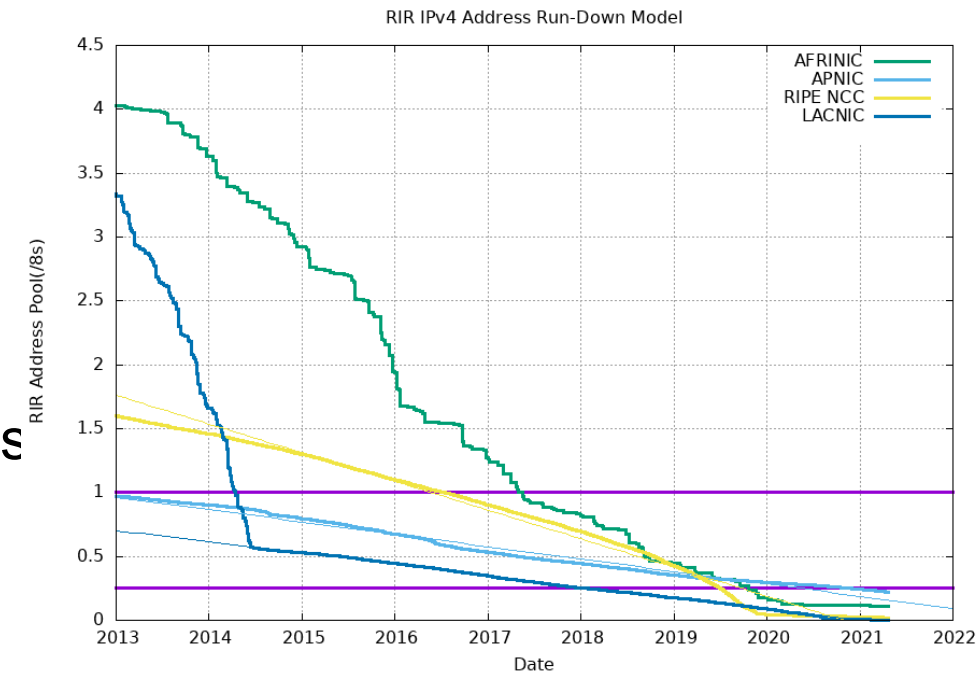
Layers	Protocols	Implemented by
Application layer	HTTP, HTTPS, FTP, DNS, ...	Application (Web browser, online game, ...) and OS
Transport layer	TCP UDP	OS (Windows, Linux, MacOS, Android, ...)
Network layer	IPv4 IPv6	OS
Datalink layer	Ethernet	Driver on OS + NIC
Physical layer		NIC

The need for IPv6

IPv4 was introduced in 1981 and deployed in 1983 for ARPANET

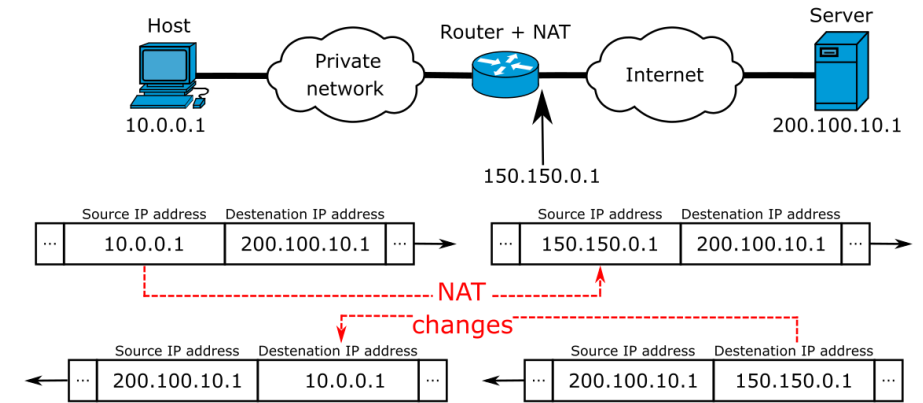
- Text based
 - Less than 600 devices connected
 - No World Wide Web, no streaming, no QoS
 - 32-bit addresses → 4.3 billion theoretical addresses
 - 7.6 billion people on earth
 - Multiple devices per person
 - IoT, cars (in 2016, more new cars than mobile phones)
- Not ideal global distribution of address blocks
- Large and complex routing tables
 - Waste of large number of addresses

NAT temporary solution to the address depletion problem



Problems with NAT

- Checksum recalculation
- ICMP manipulation like Destination host unreachable
- IPsec problems: changing IP addresses → fail integrity check
- No end-to-end reachability: port forwarding
- Decreased performance
- NAT is Not Security



IPv6

- IPv6 is designed to be the successor to IPv4
- Depletion of IPv4 address space has been the motivating factor for creating IPv6
- Introduced in 1998
- Increased address space with 128 bit addresses
 - 340,282,366,920,938,463,463,374,607,431,768,211,456 addresses
- Opportunity to improve IP protocol and processes

transition to ipv6

WHY?

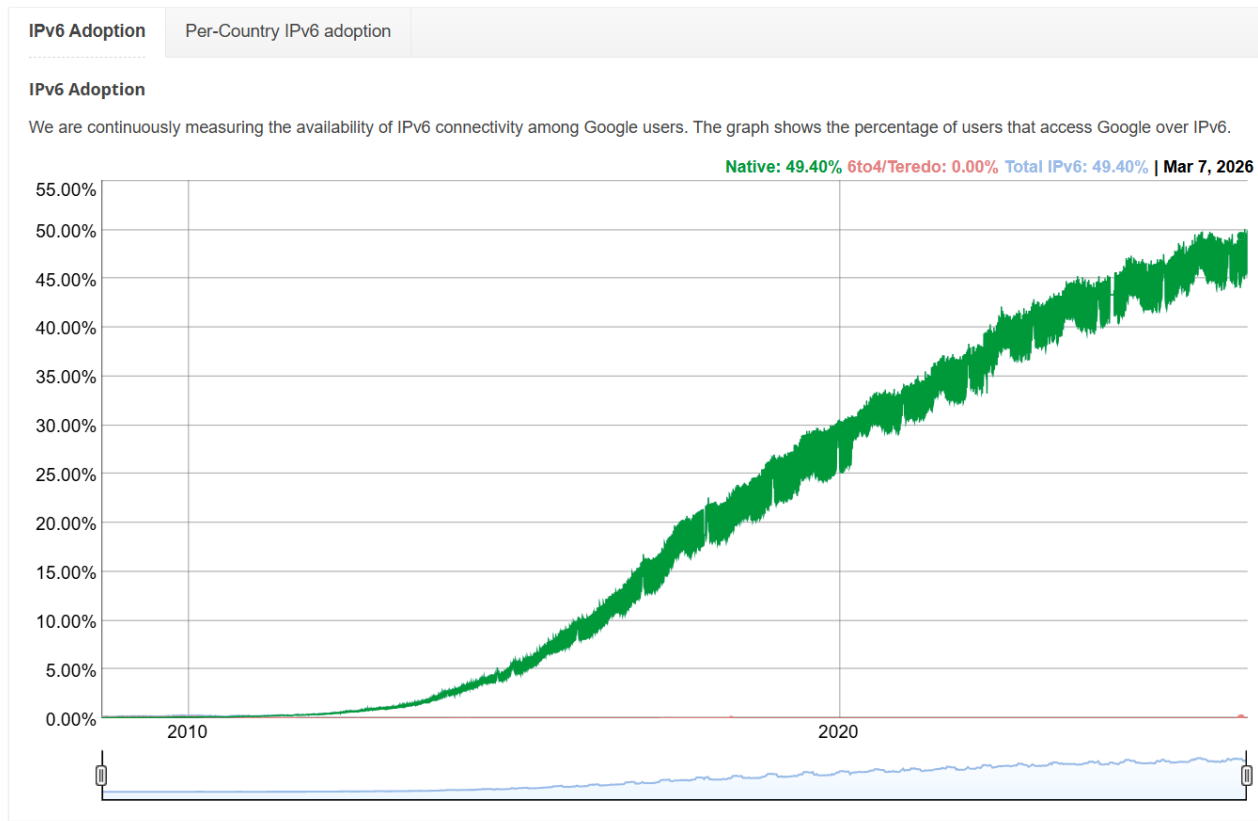
- IPv4 address depletion
- Access to IPv6-Only web services and users
- Increasing performance
- Improved security

WHEN?

- Not on a specified day (as was with IPv4 on 1 January 1983)
- IPv4 and IPv6 will coexist for many years in the future
- Transition is happening now

IPv6 adoption rate

<https://www.google.com/intl/en/ipv6/statistics.html#tab=ipv6-adoption>



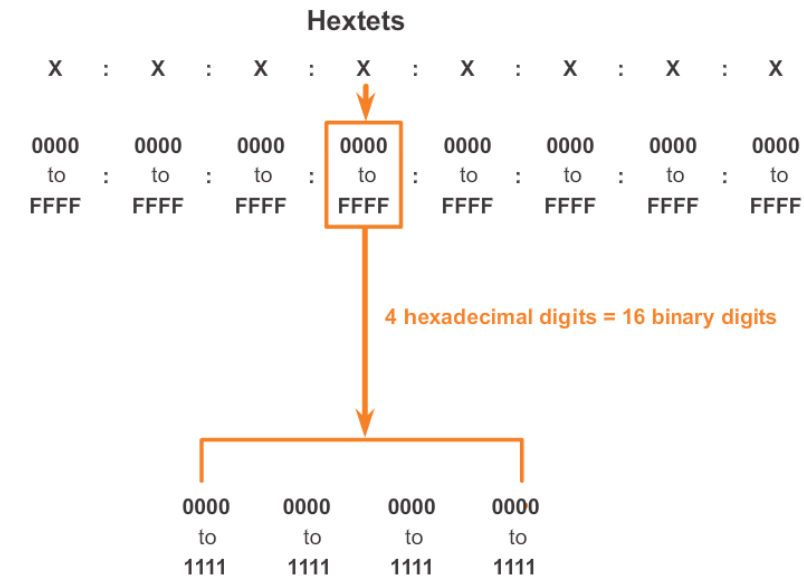
The ipv6 address

- 128 bits
- **Hexadecimal** string (4 bits = 1 hexadecimal)
- Group of 4 hexadecimal numbers = 16 bits = **hextet**
- 8 hextets (=32 total hexadecimal numbers) divided by :
- Not case sensitive
- A network interface card can have **multiple IPv6** addresses. Usually a Global Unicast and Link-local address

Examples

- 2001:0DB8:52B7:1111:0000:0000:0000:0200
- FE80:0000:0000:0000:0123:4567:89AB:CDEF
- 2A02:1810:4D1B:2100:0201:2EFF:FE65:819D
- FF02:0000:0000:0000:0000:0000:0000:0001

Hex	Binary	Hex	Binary
0	0000	8	1000
1	0001	9	1001
2	0010	A	1010
3	0011	B	1011
4	0100	C	1100
5	0101	D	1101
6	0110	E	1110
7	0111	F	1111



Rule 1 - Omitting Leading 0s

The **first rule** to help reduce the notation of IPv6 addresses is any **leading 0s** (zeros) in any 16-bit section or hextet **can be omitted**

- 01AB can be represented as 1AB
- 09F0 can be represented as 9F0
- 0A00 can be represented as A00
- 00AB can be represented as AB

Preferred	2001:0DB8:0000:1111:0000:0000:0200
No leading 0s	2001: DB8: 0:1111: 0: 0: 0: 200

Preferred	2001:0DB8:0000:A300:ABCD:0000:0000:1234
No leading 0s	2001: DB8: 0:A300:ABCD: 0: 0:1234

Preferred	2001:0DB8:000A:1000:0000:0000:0000:0100
No leading 0s	2001: DB8: A:1000: 0: 0: 0: 100

Preferred	FE80:0000:0000:0000:0123:4567:89AB:CDEF
No leading 0s	FE80: 0: 0: 0: 123:4567:89AB:CDEF

Rule 2- Omitting All 0 Segments

A double colon (::) can replace any single, **contiguous string** of one or more 16-bit segments (hexets) consisting of **all 0's**. Double colon (::) can only **be used once** within an address otherwise the address will be ambiguous. When used with the omitting leading 0s technique → the notation of IPv6 address can often be greatly reduced.

=> known as the compressed format

Incorrect address: 2001:0DB8::ABCD::1234

Possible expansions of ambiguous compressed addresses:

- 2001:0DB8::ABCD:0000:0000:1234
- 2001:0DB8::ABCD:0000:0000:0000:1234
- 2001:0DB8:0000:ABCD::1234
- 2001:0DB8:0000:0000:ABCD::1234

Preferred	FE80:0000:0000:0000:0123:4567:89AB:CDEF
No leading 0s	FE80: 0: 0: 0: 123:4567:89AB:CDEF
Compressed	FE80::123:4567:89AB:CDEF

Preferred	0000:0000:0000:0000:0000:0000:0000:0001
No leading 0s	0: 0: 0: 0: 0: 0: 0: 1
Compressed	::1

Preferred	FF02:0000:0000:0000:0000:0000:0000:0001
No leading 0s	FF02: 0: 0: 0: 0: 0: 0: 1
Compressed	FF02::1

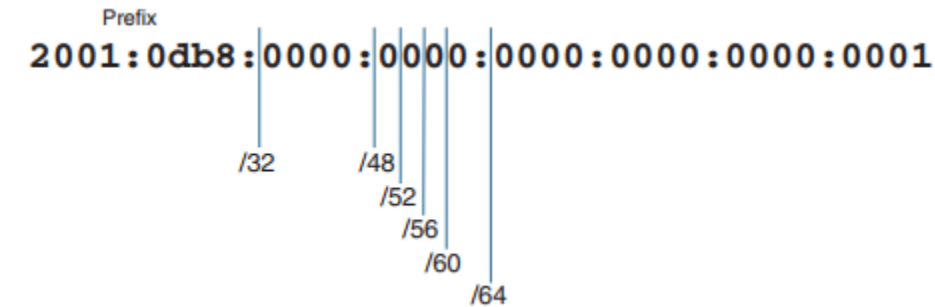
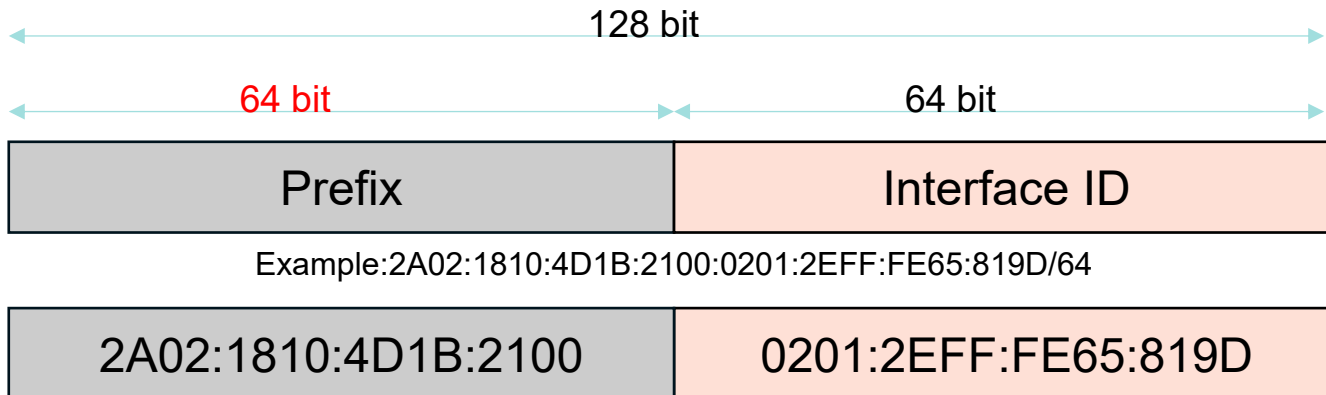
Prefix Length (like IPv4 subnet mask)

Decimal value indicating the number of leftmost bits that identify the **prefix** (equivalent to network portion of IPv4 address)

The rest of the IPv6 address is the **Interface ID** (equivalent to host portion of IPv4 address)

Notation: ipv6-address/prefix-length

Typical prefix length for end devices is **/64**



IPv6 Address Types

Unicast address

- Uniquely **identifies an interface**
- Packet send to a unicast address is received by the i
- Can be source and destination address

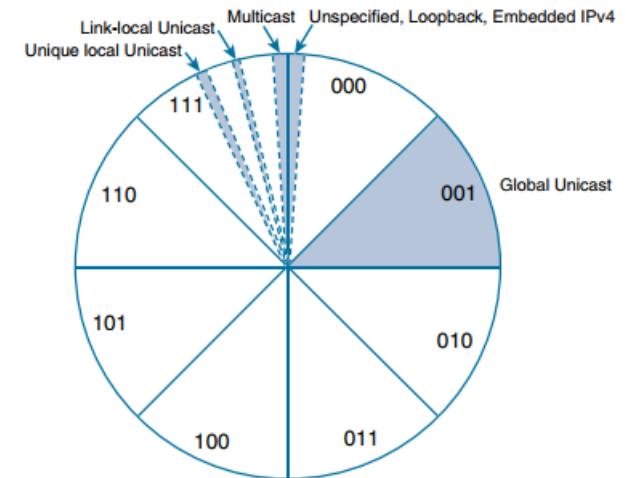
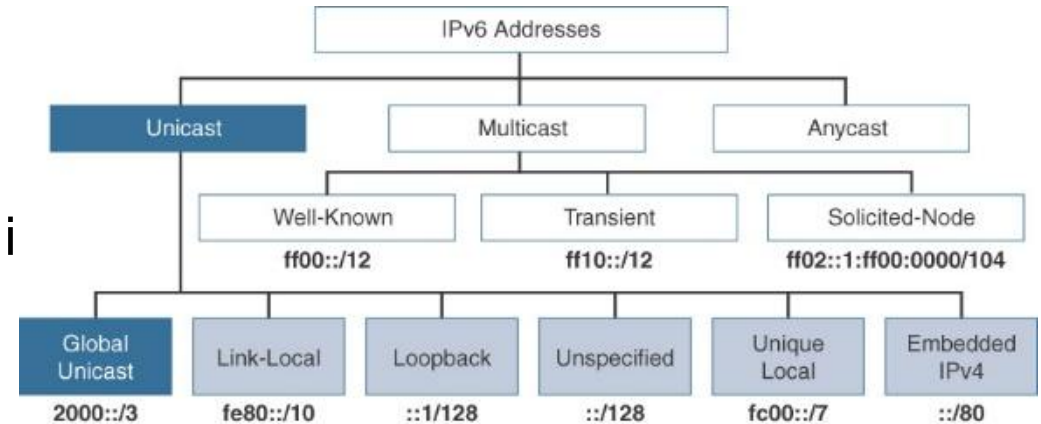
Multicast

- Identifies a multicast **group listening** to the address
- Packet send to a multicast address is received by all interface multicast group
- Can only be a destination address

Anycast

- Multiple interfaces (devices) with the same unicast address
- Packets sent to an anycast address is routed to the device wi

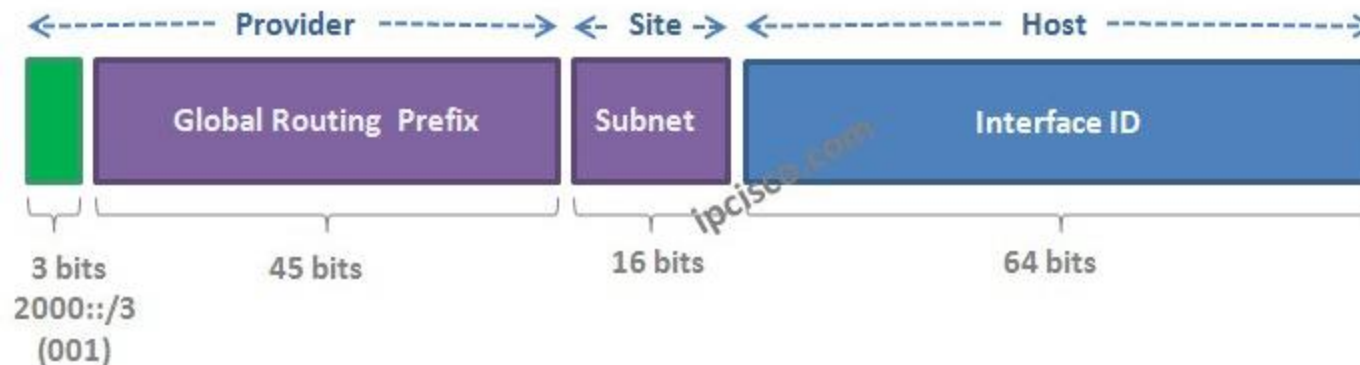
There is **no broadcast** in IPv6!



Global Unicast addresses (Unicast)

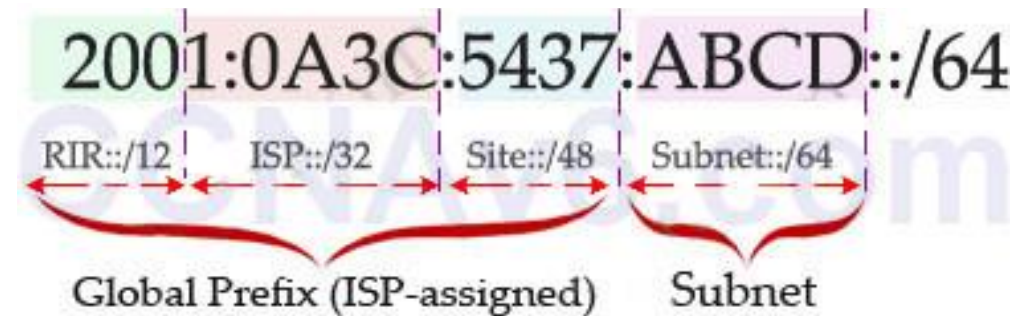
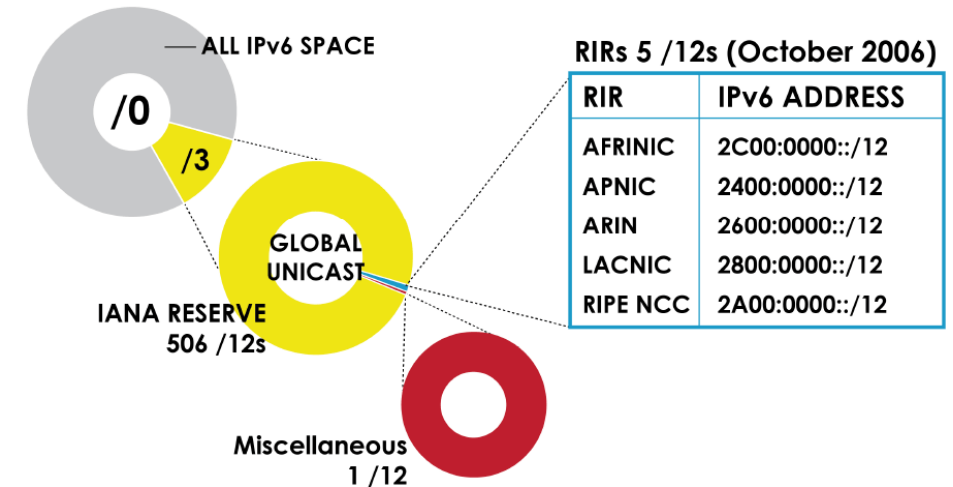
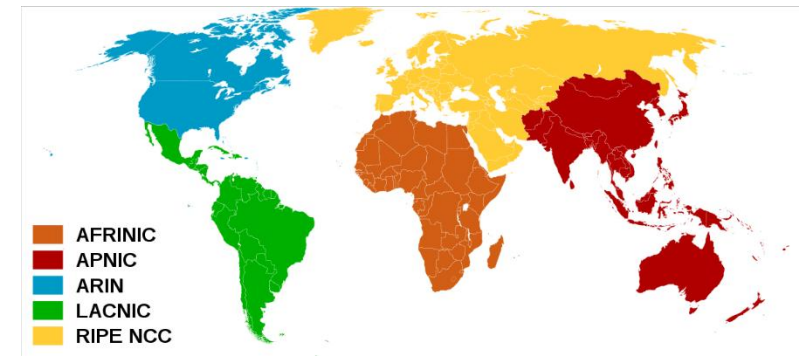
Global Unicast Address (GUA)

- routable address in the IPv6 internet
- Similar to IPv4 public IP
- In IPv6, all end devices should get a Global Unicast Address to access the internet, No NAT
- 2000::/3 addresses (first hextet from 2000 to 3fff)
 - **Global Routing Prefix:** for a specific **organization's site**. This is assigned to an individual site of an organization.
After this prefix is assigned, routers on the IPv6 Internet (with possible another prefix) will forward IPv6 traffic to routers of the organization's site
 - **Subnet ID:** used within an organization's site to create and identify **subnets**.
 - **Interface ID:** indicates the interface on a specific subnet within the site. Is equivalent to the host portion of an IPv4 address. Interface ID because a host can have multiple interfaces.



IPv6 address distribution

- Regional Internet registries (RIR) responsible for distribution of IPv6 addresses to ISPs
- Logical distribution → smaller and efficient internet routing table
- IPv4 → uneven, fragmented distribution → large and complex routing tables



Global Unicast Address configuration

Manually configured

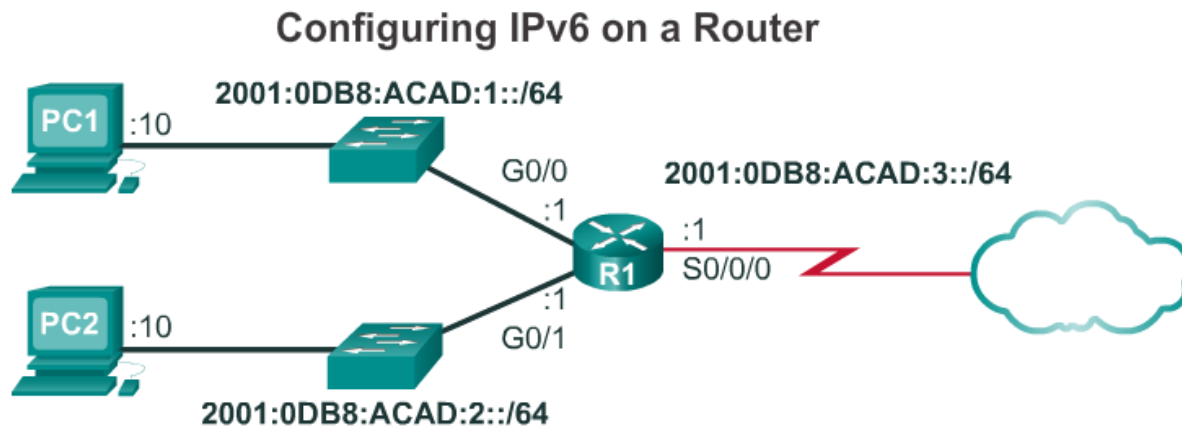
- Static IPv6 address

Stateful DHCPv6

- Full address and options configured by DHCPv6 server
- Similar to DHCP in IPv4

Stateless Address Autoconfiguration (SLAAC)

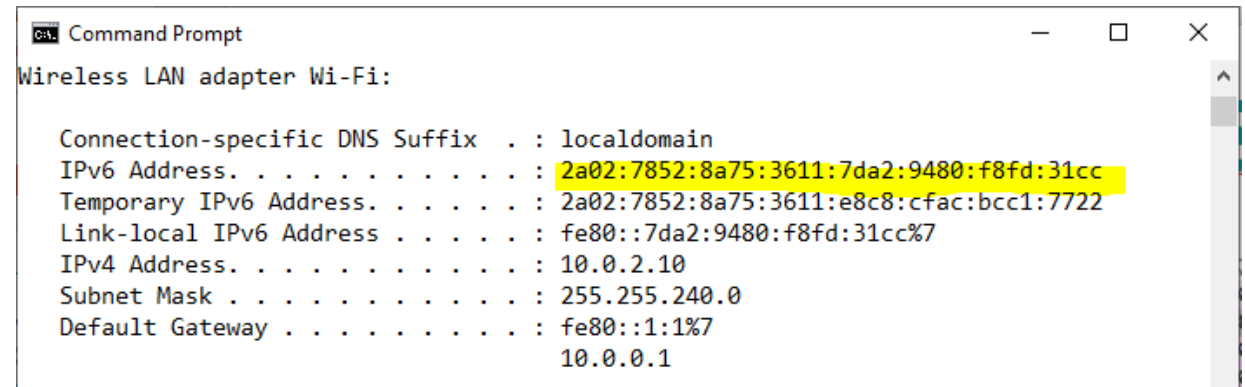
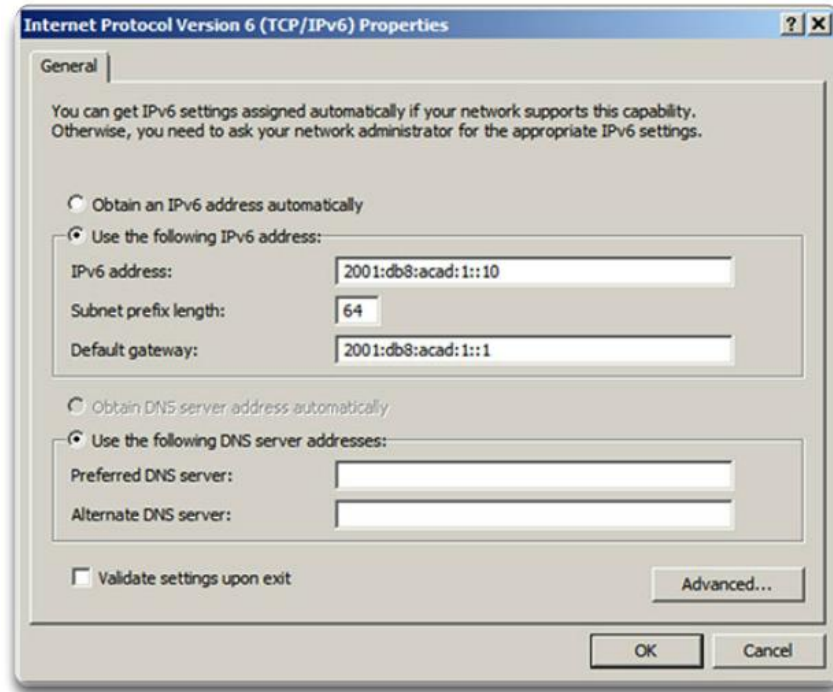
Configuration of a Global Unicast Address



```
R1(config)#interface gigabitethernet 0/0
R1(config-if)#ipv6 address 2001:db8:acad:1::1/64
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface gigabitethernet 0/1
R1(config-if)#ipv6 address 2001:db8:acad:2::1/64
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface serial 0/0/0
R1(config-if)#ipv6 address 2001:db8:acad:3::1/64
R1(config-if)#clock rate 56000
R1(config-if)#no shutdown
```


Configuration of a Global Unicast Address

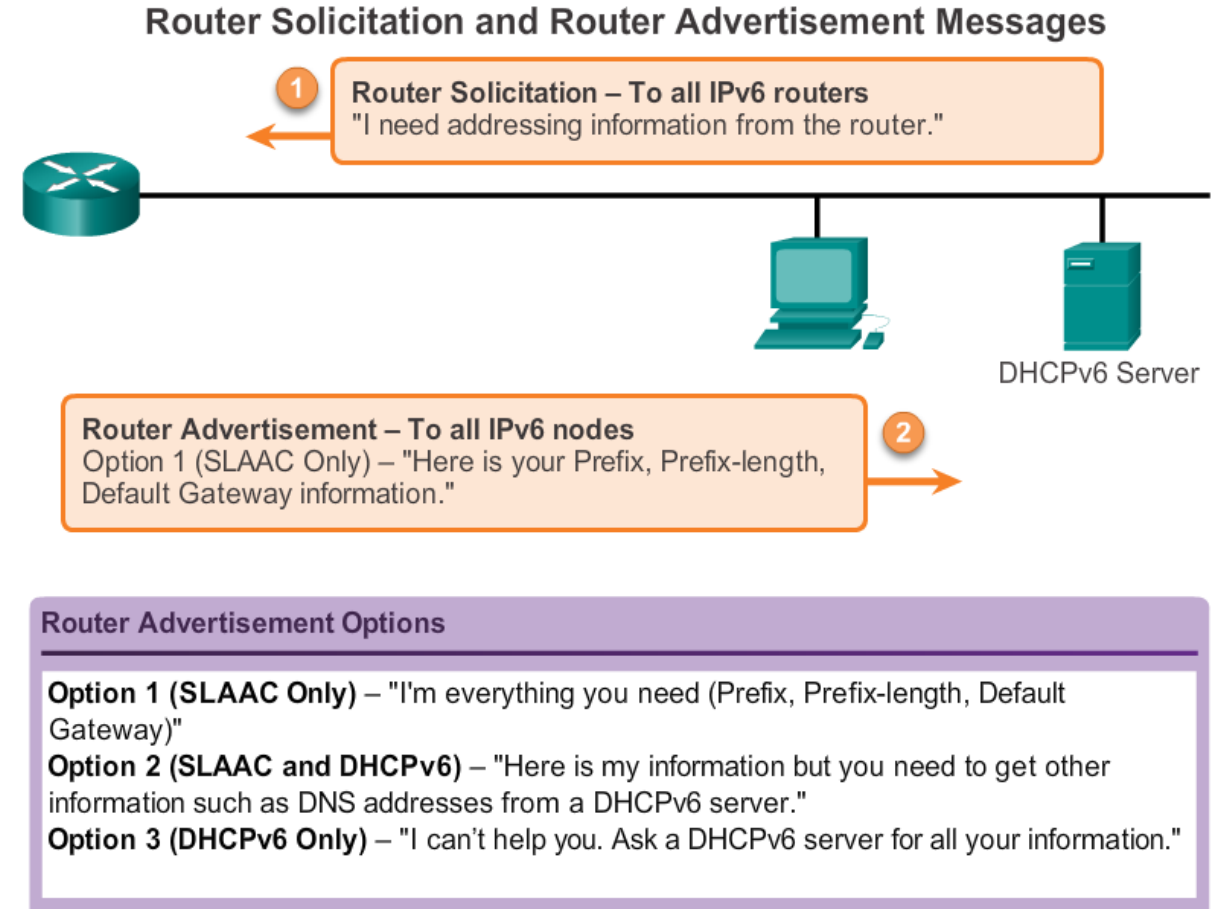
Static configuration for servers, printers, routers, ...



Stateless Address Autoconfiguration (SLAAC)

Stateless Address Autoconfiguration (SLAAC)

- obtain **prefix, prefix length, and default gateway** address information from an IPv6 router without the use of a DHCPv6 server
- Devices rely on the local router's ICMPv6 **Router Advertisement (RA)** messages to obtain the necessary information
- Interface ID is calculated by the client
- Can use a stateless DHCPv6 server for additional options (e.g. DNS server)

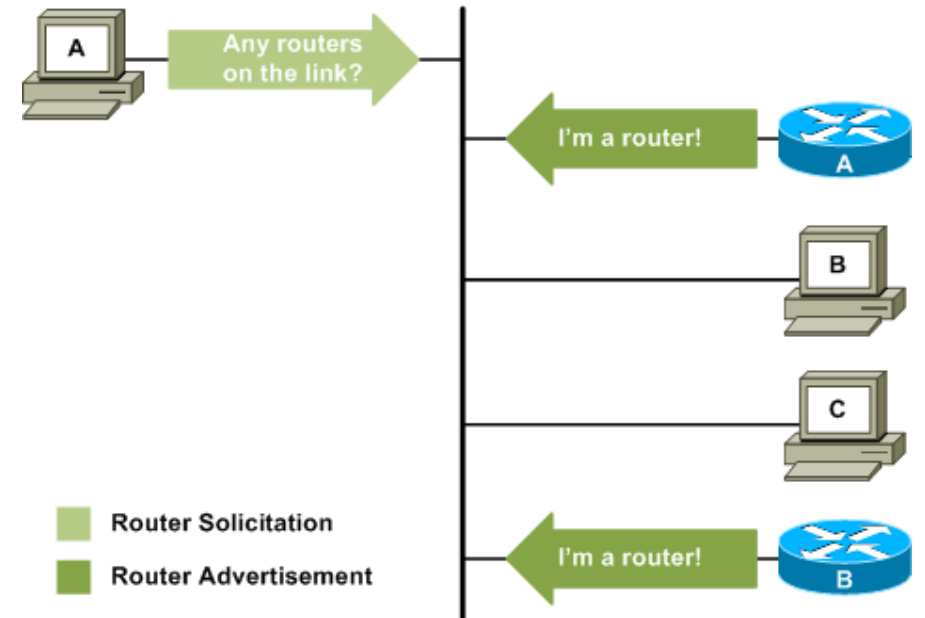


Router solicitation/ router advertisement

Router solicitation is used by hosts to get prefix, prefix length and default gateway from router

FF02::2 multicast is used as destination address

Routers send **router advertisement** periodically or as a response to a router discovery containing the prefix, prefix length and default gateway and information about DHCPv6 server
Multiple routers can send router advertisement packets



EUI-64 Process or Randomly Generated interface Id

48 bit MAC address.



Additional 16 bits

FF FE



Uniqueness of the MAC

000000X0

where

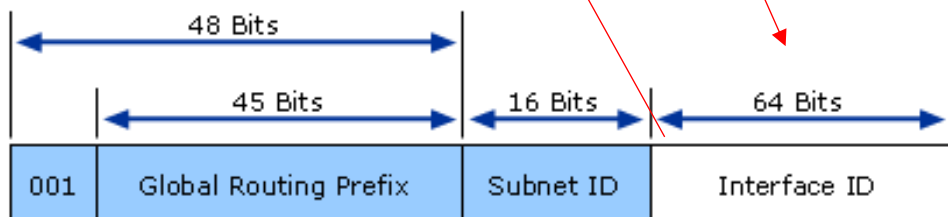
Flip 7th bit

X = 1

64 bits Eui-64 address



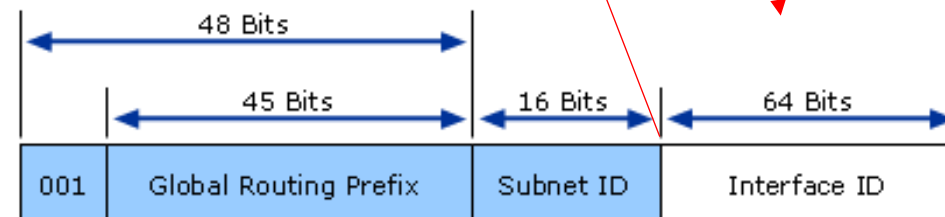
The Global Unicast Address



Global unicast addresses contain four fields.



The Global Unicast Address



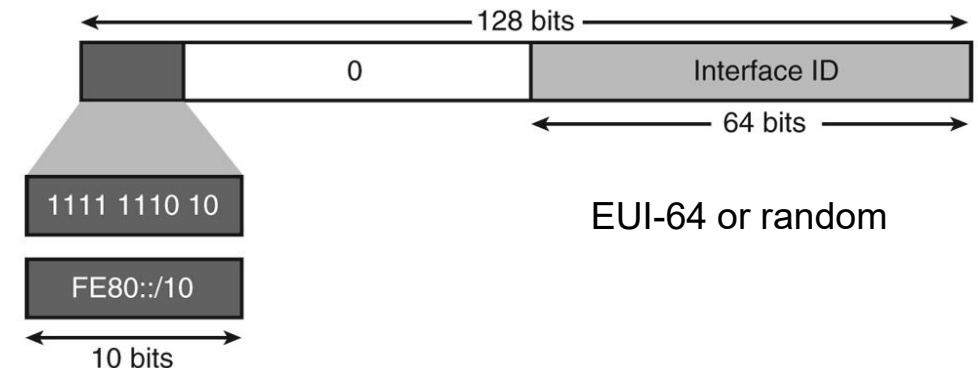
Global unicast addresses contain four fields.

Link-local unicast Address (Unicast)

- Used only to communicate with devices on the same link (subnet)
- Not routable
- Device can give itself a Link-local Unicast Address or can be manually configured
- Can be used by end devices to communicate with router or DHCPv6 server to obtain global unicast address or by routers for dynamic routing protocols (EIGRP, OSPFv3)
- Each interface must have a Link-Local Unicast Address
- FE80::/10 (first hextet from FE80 to FEBF)

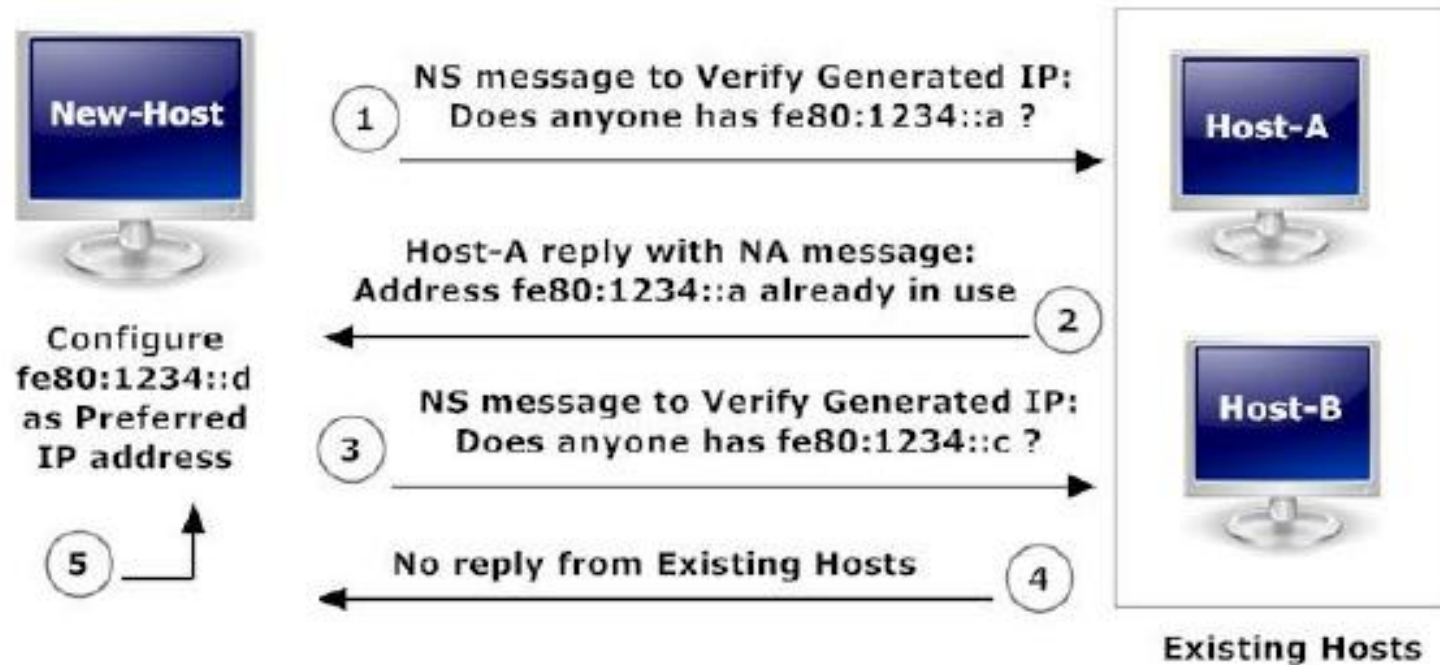
```
Command Prompt
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : localdomain
IPv6 Address. . . . . : 2a02:7852:8a75:3611:7da2:9480:f8fd:31cc
Temporary IPv6 Address. . . . . : 2a02:7852:8a75:3611:e8c8:cfac:bcc1:7722
Link-local IPv6 Address . . . . . : fe80::7da2:9480:f8fd:31cc%7
IPv4 Address. . . . . : 10.0.2.10
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . : fe80::1:1%7
                          10.0.0.1
```



IPv6 Duplicate Address Detection (DAD)

ICMPv6 Neighbor Solicitation messages (similar to IPv4 ARP request)



Other Unicast addresses

Loopback address
::1/128



- Similar to 127.0.0.1 IPv4 address
- To test the TCP/IP stack or to access local services

Unspecified address
::/128



- Can be used as a source address to obtain an IPv6 address
- Can never be assigned to interface or used as a destination address

Unique local address
FC00::/7

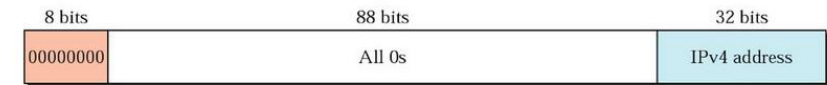
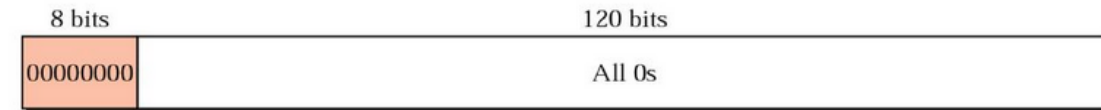
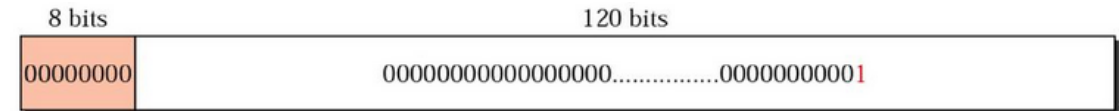


- Similar to IPv4 private ranges

Embedded IPv4
::/80



- Used to communicate with IPv4 network



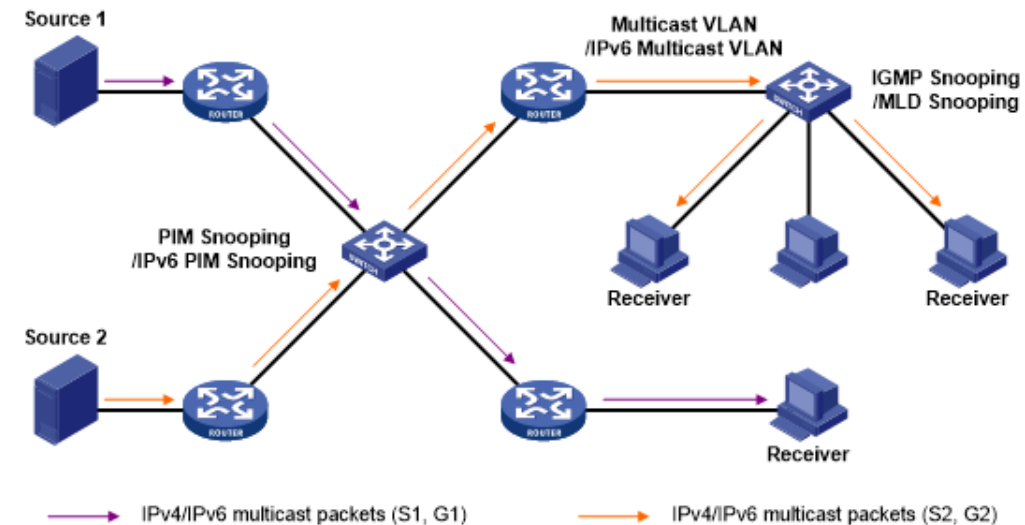
a. Compatible address



b. An example of address transformation

Multicast

- **One-to-many** (Broadcast = one to all)
- Device that wants to receive a multicast **subscribes** to a specific multicast address = multicast receiver
- The receiver sends an MLD (IPv6) or IGMP (IPv4) packet on the network informing the network devices it wants to listen to a specific multicast address
- A multicast sender sends the multicast packets on the network
- Network devices will only deliver the multicast to the subscribed devices



Multicast

More efficient than broadcast because it can be “filtered” by switches and routers

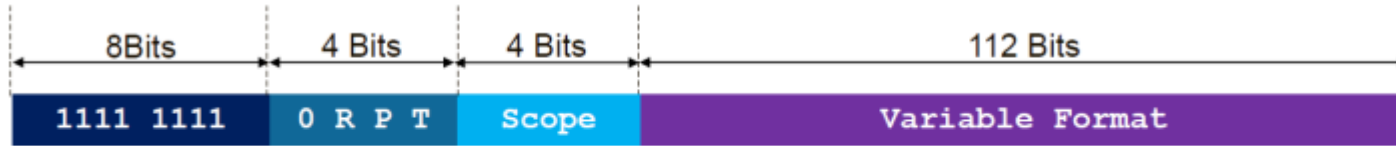
- Internet Group Management Protocol (IGMP) IPv4
- Multicast Listener Discovery (MLD) IPv6

Prefix of FF00::/8 (first octet from FF00 to FFFF)

Multicast addresses can only be a destination address

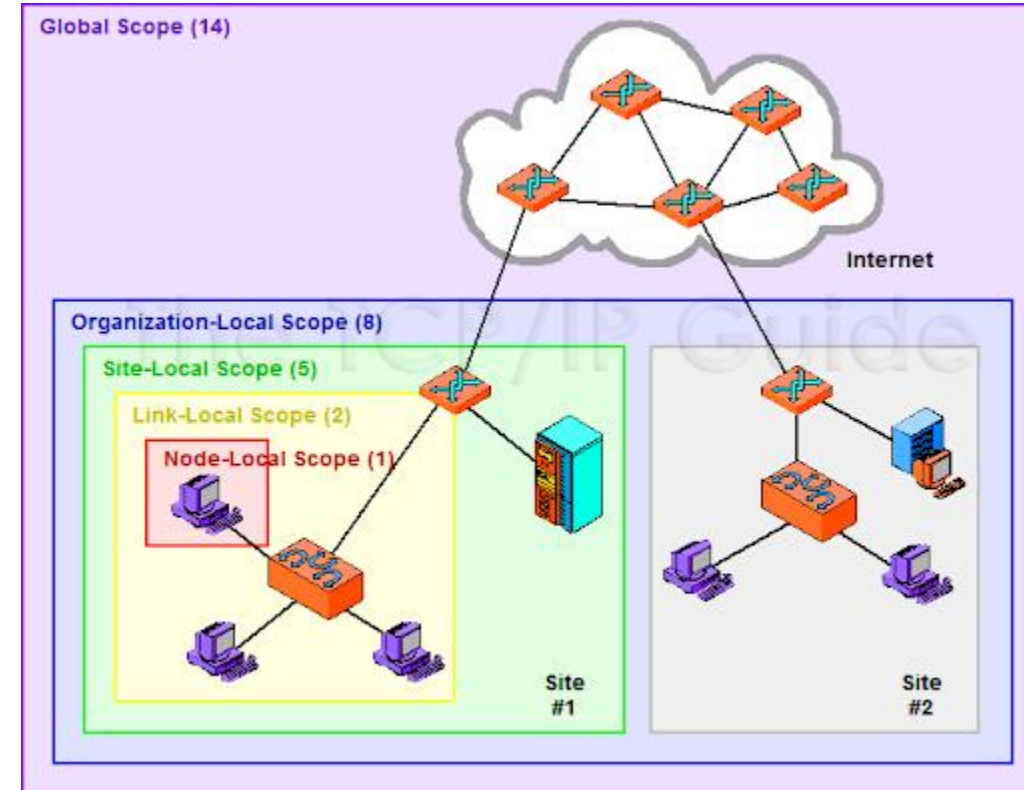
- To send to multicast
 - add multicast address as destination
- To receive multicast
 - register multicast MAC address with network adapters
 - Send MLD packet to inform routers and switches that you want to receive a multicast

Multicast addresses



Flags	
R = 0	No embedded RP
R = 1	Embedded RP
P = 0	Not based on unicast
P = 1	Based on unicast
T = 0	Permanent address (IANA assigned)
T = 1	Temporary address (local assigned)

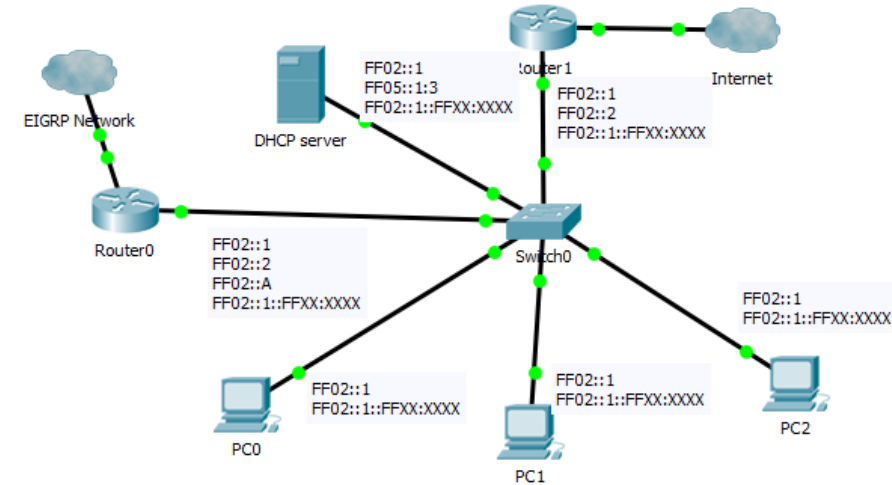
Scope	
1	Node
2	Link
3	Subnet
4	Admin
5	Site
8	Organisation
E	Global



Well known multicast addresses

Permanent reserved addresses

Address	Listeners
FF02::1	All nodes on subnet
FF02::2	All routers on subnet
FF02::5	OSPFv3 routers on subnet
FF02::A	EIGRP routers on subnet
FF05::1:3	All DHCP servers on site
FF02::1:FFXX:XXXX	Solicited-Node



```
Router1#show ipv6 interface gigabitEthernet 0/1
GigabitEthernet0/1 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::1
No Virtual link-local address(es):
Global unicast address(es):
  2001:DB8:2000::1, subnet is 2001:DB8:2000::/64
Joined group address(es):
  FF02::1
  FF02::2
  FF02::1:FF00:1
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
```

Solicited-node multicast

- More efficient process than broadcast
- Used for **Neighbor Discovery (ND)**, similar to ARP in IPv4 and **Duplicate Address Detection (DAD)**
- All unicast addresses are mapped to solicited-node multicast
- Each unicast address generates a solicited-node multicast
- Possible that more than one device has the same solicited-node multicast
 - Upper layer will examine the destination IPv6 address and discard the packet if there is not a match

IPv6 Unicast Address

2001:0DB8:CAFE:0001:0000:0000:ED45:034A

Copy 24 bits

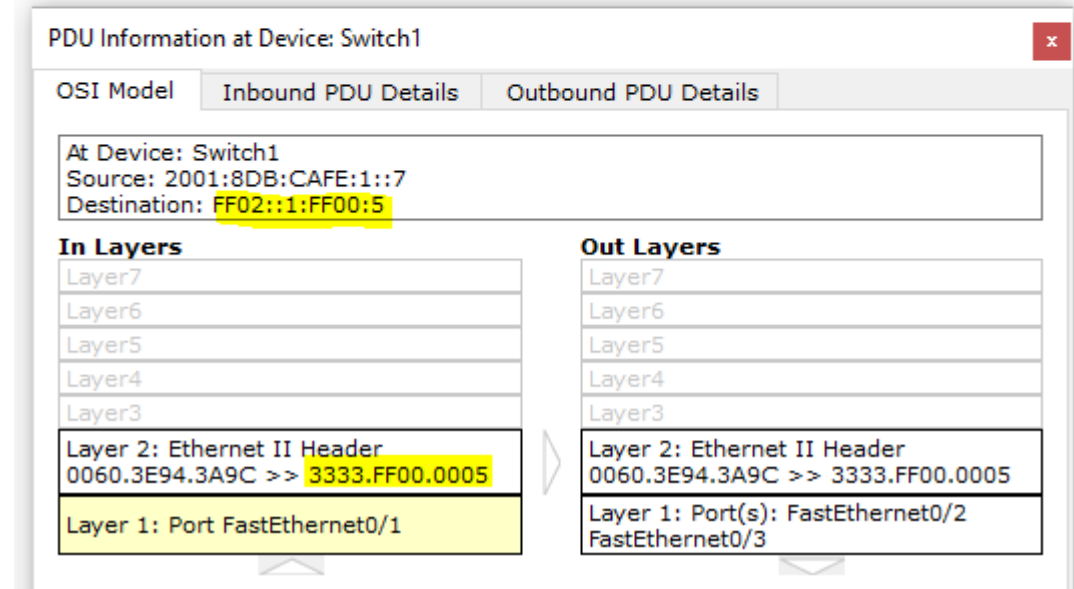
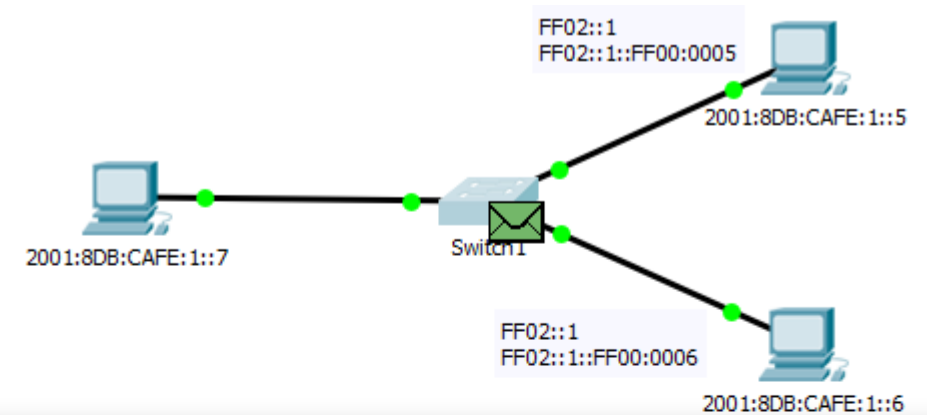
Solicited-node Multicast Address

FF02:0000:0000:0000:0000:0001:FF45:034A

```
Router1#show ipv6 interface gi 0/1
GigabitEthernet0/1 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::2D0:FFFF:FEAE:DE02
No Virtual link-local address(es):
Global unicast address(es):
  2001:DB8:2000::1, subnet is 2001:DB8:2000::/64
Joined group address(es):
  FF02::1
  FF02::2
  FF02::1:FF00:1
  FF02::1:FFAE:DE02
MTU is 1500 bytes
```

Neighbor discovery protocol (NDP)

- A **Neighbor Solicitation** packet is used by devices to get MAC address of a unicast IPv6 address
 - Similar to ARP in IPv4 broadcast
- **Solicited-node multicast** is added as destination address
- Only NIC's listening to this multicast send the frame to the upper layer
- Switches with MLD snooping enabled filter packets
- Host with correct IPv6 address answers with **Neighbor Advertisement** containing its MAC address



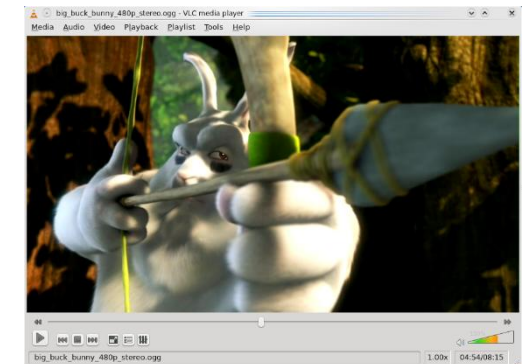
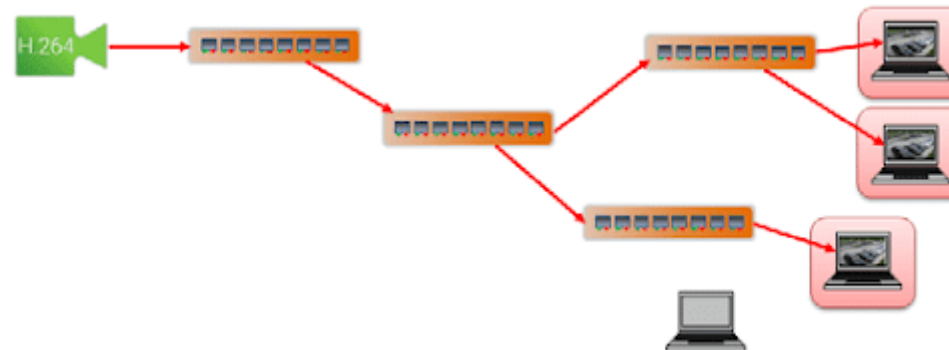
Transient multicast (dynamic multicast)

FF10::/12 address range

Used as temporary multicast addresses by applications

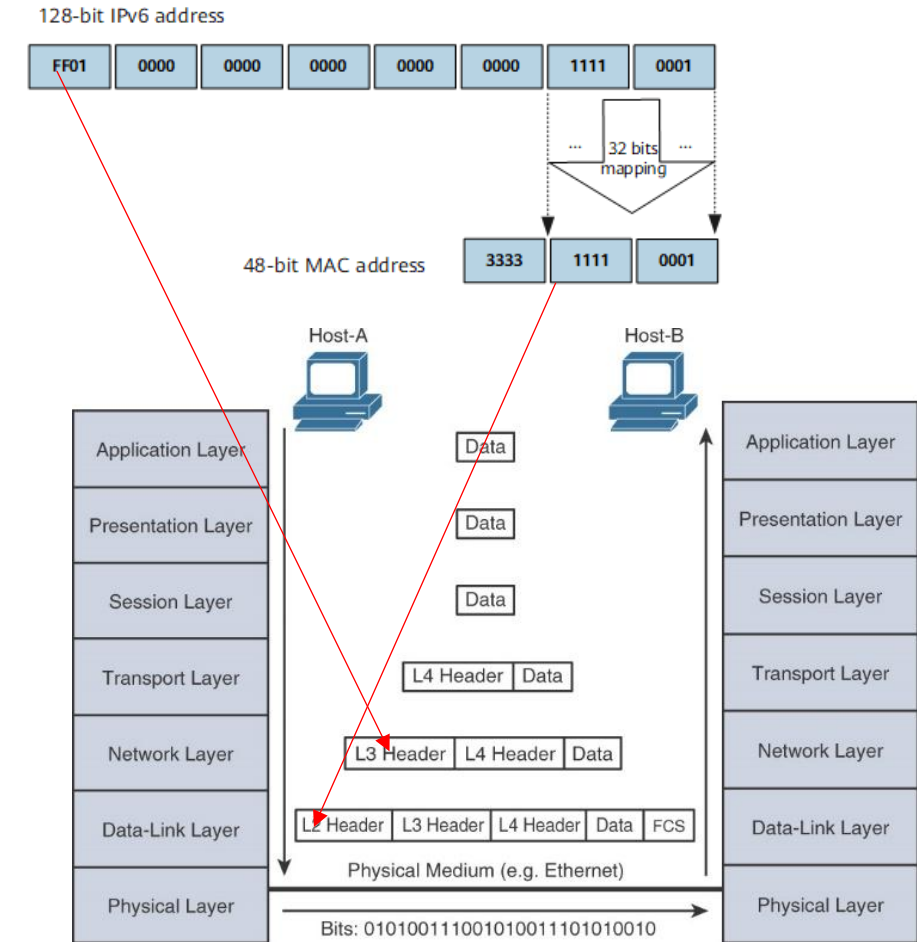
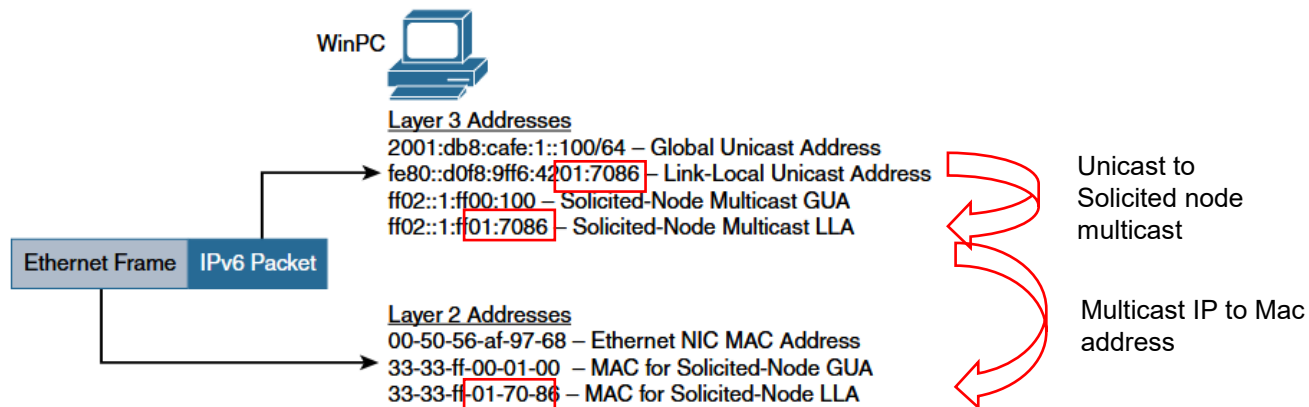
- Audio/Video streaming
- Video Conferencing
- OS deployment

Multicast Listener Discovery (MLD) is used by hosts to subscribe to or leave a multicast group



Mapping multicast to MAC address

- NIC's listen to MAC addresses
- When a device wants to listen to a multicast, a multicast MAC address needs to be added to the MAC addresses to which the NIC is listening to
- A device sending to a multicast group (adding the multicast group IP address), will add the multicast MAC address on the datalink layer
- IPv6 multicast MAC address starts with 3333, the last 32 bit of the IPv6 multicast address is mapped to the MAC address



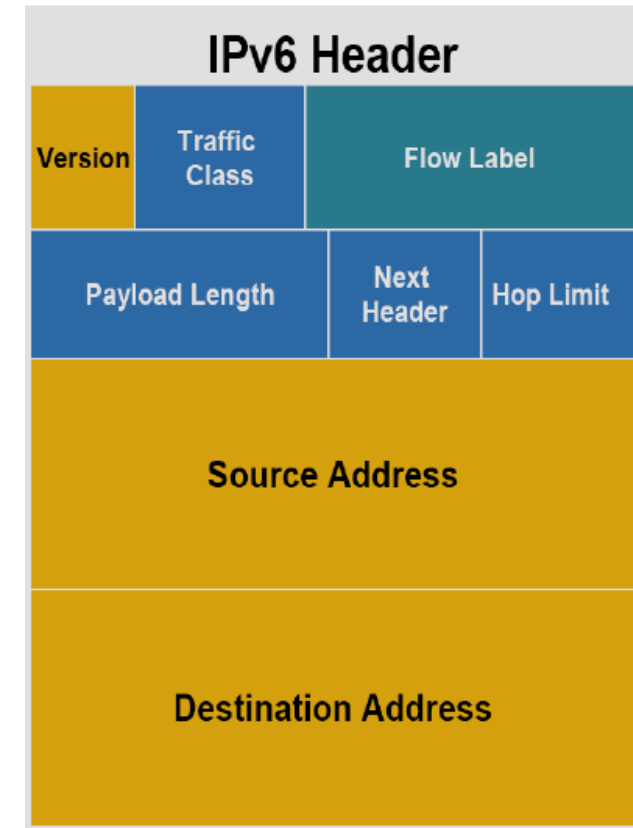
Overview IPv6 Addresses

First <u>Hextet</u> (Far Left)	Type of IPv6 Address
0000 to 00FF	Loopback address, any address, unspecified address, or IPv4-compatible
2000 to 3FFF	Global unicast address (a routable address in a range of addresses that is currently being handed out by the Internet Assigned Numbers Authority [IANA])
FE80 to FEBF	Link-local (a unicast address which identifies the host computer on the local network)
FC00 to FCFF	Unique-local (a unicast address which can be assigned to a host to identify it as being part of a specific subnet on the local network)
FF00 to FFFF	Multicast address

IPv6 header

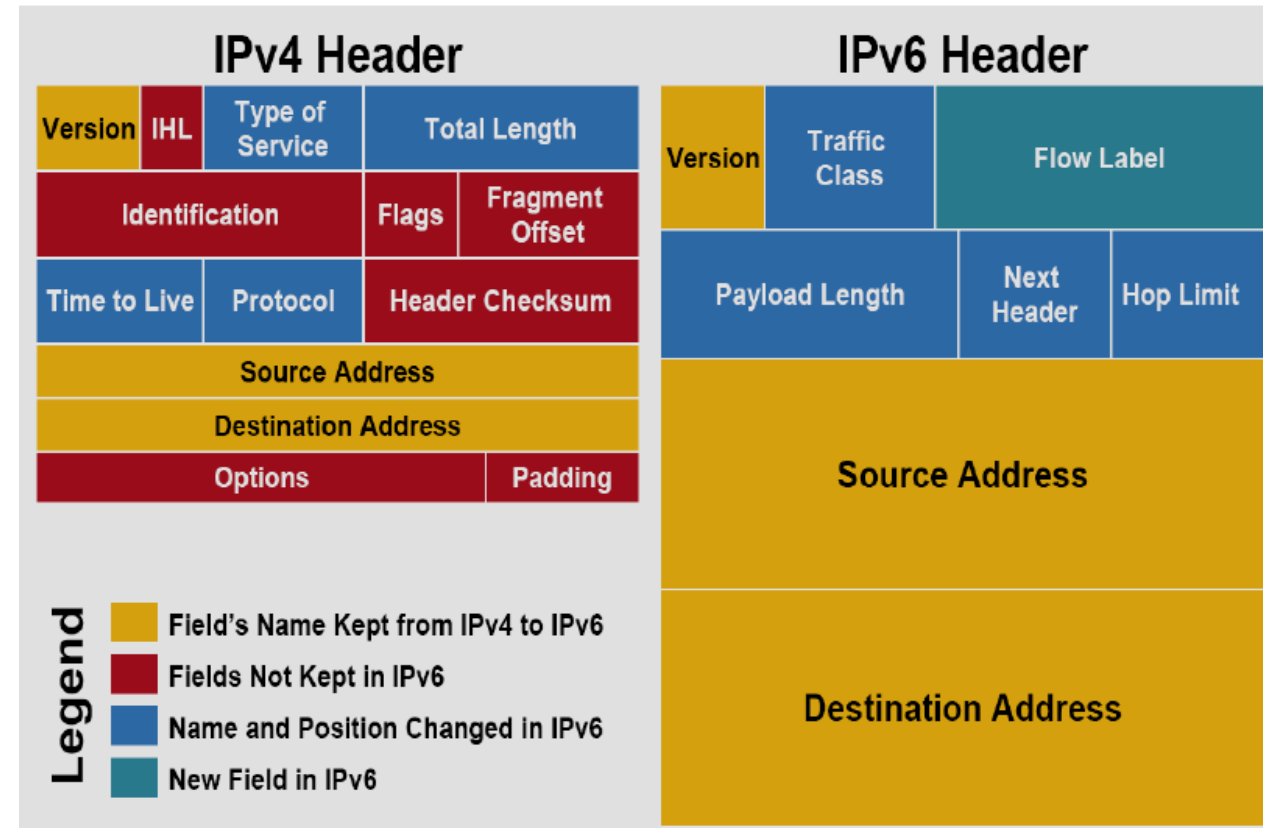
Simplified header with fixed 40 byte header improves packets handling

- Version: 6
- Traffic Class: Quality of Service
- Flow Label: Identify packets in the same flow
- Payload Length: length of payload + extension headers. 2 byte = max 65 536 bytes
- Next Header: Upper layer protocol or extension header
- Hop Limit: Decrement by every router to prevent a packet from eternally staying on the network
- Source and Destination Address: 128 bit IPv6 address



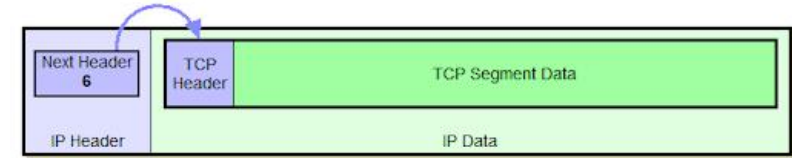
IPv6 header vs IPv4 header

- IP Header Length: not needed because IPv6 has a 40 byte header size
- Identification, Flags, Fragment Offset: Routers do not fragment packets as they do in IPv4
- Header checksum: Ethernet, TCP and UDP have checksums. IP is best effort
- Options: replaced by extension headers
- Padding: not needed, fixed size header

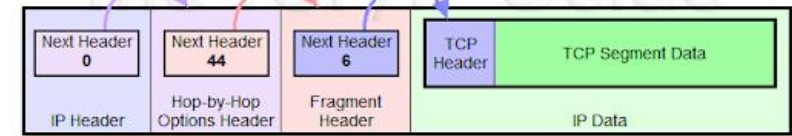


Extension headers

- Options are separated from fixed size main header
- Additional headers can add information and functionality to the IPv6 packet
- Extension headers are only examined by the receiver, except the Hop-by-hop extension header
- All extension headers have a “Next Header” field
- An IPv6 packet can have none, one or many extension headers
- Future proof, new headers can be defined and added



IPv6 Datagram With No Extension Headers Carrying TCP Segment

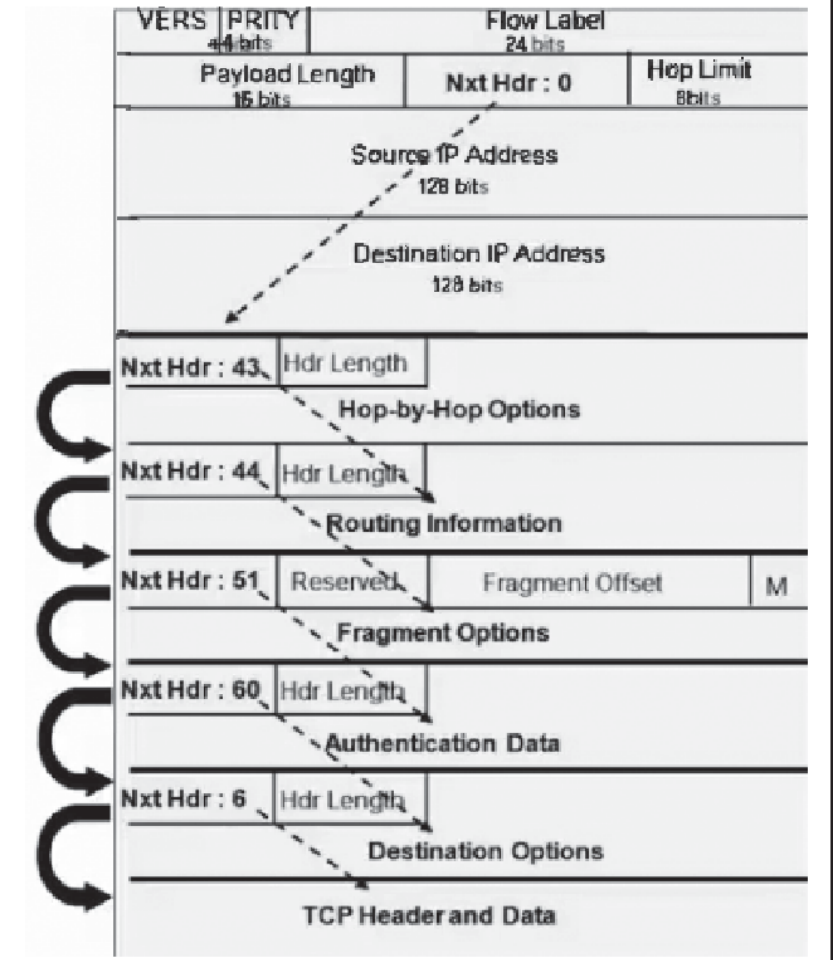


IPv6 Datagram With Two Extension Headers Carrying TCP Segment

Code	Next Header
0	Hop-by-hop option
2	ICMPv6
6	TCP
17	UDP
44	Fragmentation
50	Encrypted security payload
51	Authentication
60	Destination option

Extension headers

- Hop-by-hop: header inspected by every hop
 - E.g. jumbograms to extend maximum packet size
- Fragmentation: source fragments packet if MTU of a link on the path is too small
- Authentication: Validates sender and data integrity (IPsec)
- Encrypted security payload: confidentiality (IPsec)

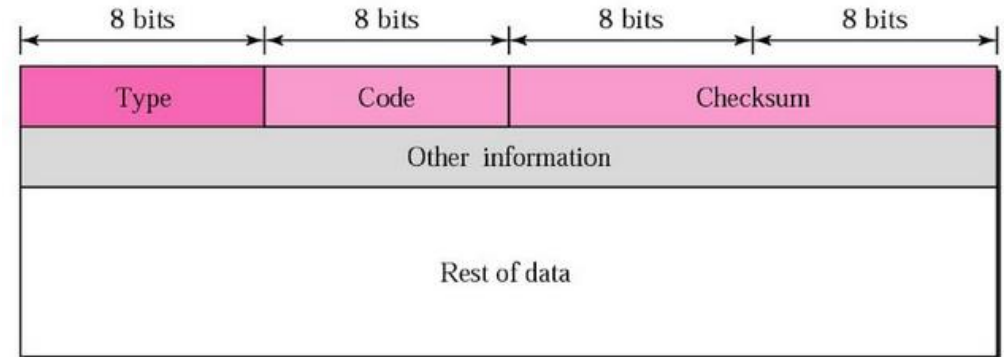


ICMPv6

ICMPv6 combines different functionalities of IPv4 (ICMP, IGMP, ARP)

Error reporting messages

- Destination unreachable
- Packet too big
- Time exceeded
- Parameter problem



ICMPv6

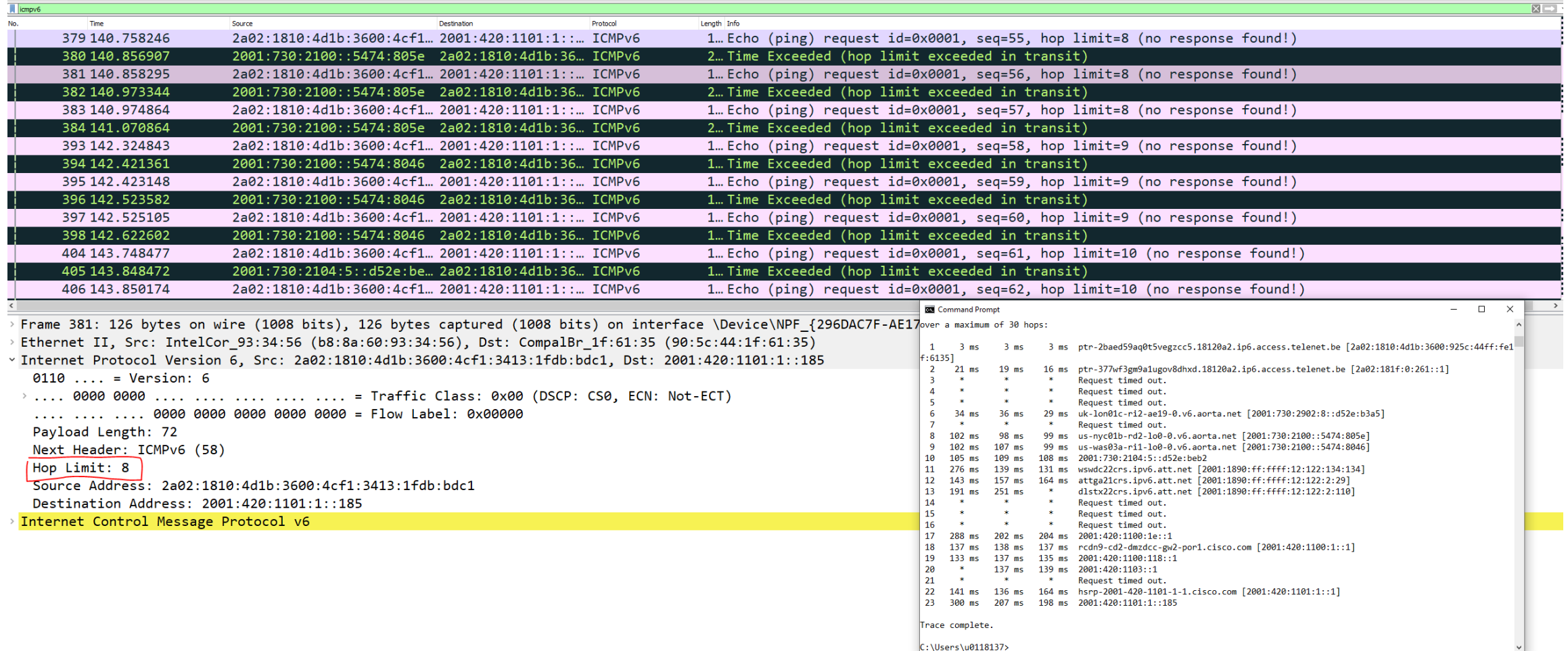
Query messages

- **Echo request and reply** (ping)
- **Control messages** (Hop limit exceeded, ...)
- **Router solicitation**: used by hosts to find routers in the network, multicast
- **Router advertisement**: send by routers (periodically or response) that contain router information (prefix, default gateway, DHCPv6, ...)
- **Neighbor Solicitation**: similar to IPv4 ARP protocol, multicast
- **Neighbor advertisement**: response to neighbor solicitation
- **Multicast Listener Discovery**: subscribe or leave multicast group

Tracert

- The `tracert <ipv4-address>` or `tracert -6 <ipv6 address>` commands will list all hops between your device and the target device.
- A ping (ICMP) will be send with a hop limit of 1
- The first router will decrease the hop limit to 0, drop the packet and send a Hop Limit Exceeded (ICMP) to the source IP address
- A ping will be send again but the hop limit will be increased by one.
- The first router will decrease the hop limit to 1, the second router will decrease the hop limit to 0, drop the packet and send a Hop Limit Exceeded to the source IP address
- This process is repeated until the destination is reached

Tracert in wireshark



The image displays a Wireshark packet capture of ICMPv6 Echo (ping) requests. The packet list shows several requests from source 2a02:1810:4d1b:3600:4cf1... to destination 2001:420:1101:1::185. The packet details for frame 381 are expanded, showing the ICMPv6 header with a Hop Limit of 8. The packet bytes pane shows the raw data of the ping request.

Frame 381: 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface \Device\NPF_{296DAC7F-AE17} over a maximum of 30 hops:

Ethernet II, Src: IntelCor_93:34:56 (b8:8a:60:93:34:56), Dst: CompalBr_1f:61:35 (90:5c:44:1f:61:35)

Internet Protocol Version 6, Src: 2a02:1810:4d1b:3600:4cf1:3413:1fdb:bdc1, Dst: 2001:420:1101:1::185

0110 = Version: 6

.... 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)

.... 0000 0000 0000 0000 = Flow Label: 0x00000

Payload Length: 72

Next Header: ICMPv6 (58)

Hop Limit: 8

Source Address: 2a02:1810:4d1b:3600:4cf1:3413:1fdb:bdc1

Destination Address: 2001:420:1101:1::185

Internet Control Message Protocol v6

Command Prompt

```
over a maximum of 30 hops:
  0  3 ms      3 ms      3 ms      ptr-2baed59aqt5vegzc5.18120a2.ip6.access.telenet.be [2a02:1810:4d1b:3600:925c:44ff:fe1f:6135]
  1  21 ms     19 ms     16 ms     ptr-377wf3gm9alugov8dhxd.18120a2.ip6.access.telenet.be [2a02:181f:0:261::1]
  2  *         *         *         Request timed out.
  3  *         *         *         Request timed out.
  4  *         *         *         Request timed out.
  5  *         *         *         Request timed out.
  6  34 ms     36 ms     29 ms     uk-lon01c-r12-ae19-0.v6.aorta.net [2001:730:2902:8::d52e:b3a5]
  7  *         *         *         Request timed out.
  8  102 ms    98 ms     99 ms     us-nyc01b-rd2-lo0-0.v6.aorta.net [2001:730:2100::5474:805e]
  9  102 ms    107 ms    99 ms     us-was03a-r11-lo0-0.v6.aorta.net [2001:730:2100::5474:8046]
 10  105 ms    109 ms    108 ms    2001:730:2104:5::d52e:beb2
 11  276 ms    139 ms    131 ms    wswdc22crs.ipv6.att.net [2001:1890:ff:ffff:12:122:134:134]
 12  143 ms    157 ms    164 ms    attga21crs.ipv6.att.net [2001:1890:ff:ffff:12:122:2:29]
 13  191 ms    251 ms     *         d1stx22crs.ipv6.att.net [2001:1890:ff:ffff:12:122:2:110]
 14  *         *         *         Request timed out.
 15  *         *         *         Request timed out.
 16  *         *         *         Request timed out.
 17  288 ms    202 ms    204 ms    2001:420:1100:1e::1
 18  137 ms    138 ms    137 ms    rcdn9-cd2-dmzdcg-gw2-por1.cisco.com [2001:420:1100:1::1]
 19  133 ms    137 ms    135 ms    2001:420:1100:118::1
 20  *         137 ms    139 ms    2001:420:1103::1
 21  *         *         *         Request timed out.
 22  141 ms    136 ms    164 ms    hsrp-2001-420-1101-1-1.cisco.com [2001:420:1101:1::1]
 23  300 ms    207 ms    198 ms    2001:420:1101:1::185

Trace complete.

C:\Users\u0118137>
```


ICMPv6 wireshark

No.	Time	Source	Destination	Protocol	Length	Info
6...	11.447...	fe80::7da2:9480:f8fd:31cc	ff02::2	ICMPv6	70	Router Solicitation from b8:8a:60:93:34:56
6...	11.451...	fe80::1:1	fe80::7da2:9480:f...	ICMPv6	2...	Router Advertisement from 00:01:2e:65:81:9f
6...	11.451...	fe80::7da2:9480:f8fd:31cc	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
6...	11.455...	fe80::7da2:9480:f8fd:31cc	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
6...	11.486...	fe80::7da2:9480:f8fd:31cc	ff02::1:ff01:1	ICMPv6	86	Neighbor Solicitation for fe80::1:1 from b8:8a:60:93:34:56
6...	11.488...	fe80::1:1	fe80::7da2:9480:f...	ICMPv6	86	Neighbor Advertisement fe80::1:1 (rtr, sol, ovr) is at 00:01:2e:65:81
6...	11.499...	fe80::7da2:9480:f8fd:31cc	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
6...	11.507...	fe80::7da2:9480:f8fd:31cc	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
6...	11.578...	fe80::7da2:9480:f8fd:31cc	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
6...	11.579...	fe80::7da2:9480:f8fd:31cc	ff02::16	ICMPv6	90	Multicast Listener Report Message v2

> Frame 635: 206 bytes on wire (1648 bits), 206 bytes captured (1648 bits) on interface \Device\NPF_{296DAC7F-AE17-483C-A781-56E5ACA9E329}, id 0

> Ethernet II, Src: PCPartne_65:81:9f (00:01:2e:65:81:9f), Dst: IntelCor_93:34:56 (b8:8a:60:93:34:56)

> Internet Protocol Version 6, Src: fe80::1:1, Dst: fe80::7da2:9480:f8fd:31cc

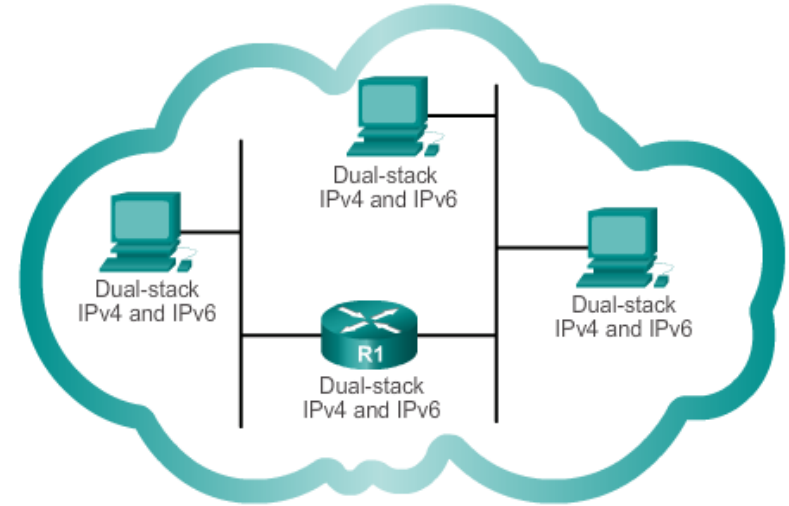
▼ Internet Control Message Protocol v6

- Type: Router Advertisement (134)
- Code: 0
- Checksum: 0x5b2e [correct]
- [Checksum Status: Good]
- Cur hop limit: 64
- > Flags: 0x00, Prf (Default Router Preference): Medium
- Router lifetime (s): 60
- Reachable time (ms): 0
- Retrans timer (ms): 0
- > ICMPv6 Option (Prefix information : 2a02:7852:8a75:3611::/64)
- > ICMPv6 Option (Route Information : Medium ::/0)
- > ICMPv6 Option (Recursive DNS Server 2620:119:53::53 2620:119:35::35)
- > ICMPv6 Option (DNS Search List Option localdomain)
- > ICMPv6 Option (MTU : 1500)
- > ICMPv6 Option (Source link-layer address : 00:01:2e:65:81:9f)

Internet Control Message Protocol v6: ProtocolPackets: 2869 · Displayed: 75 (Database.kdb - KeePass)Profile: Default

IPv4 and IPv6 Coexistence

Dual-stack: Allows IPv4 and IPv6 to coexist on the same network. Devices run both IPv4 and IPv6 protocol stacks simultaneously.

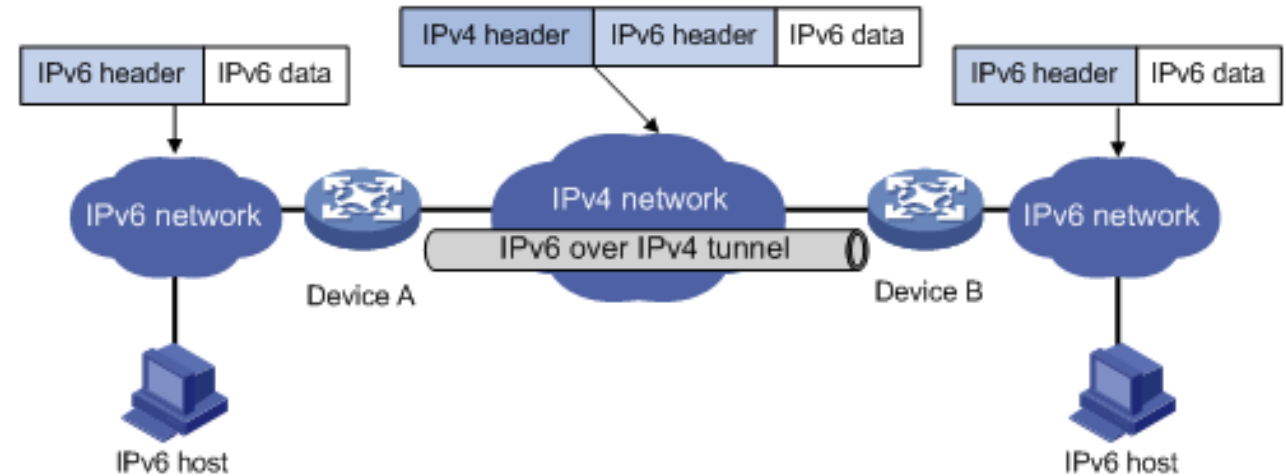


```
Command Prompt
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . : localdomain
IPv6 Address. . . . . : 2a02:7852:8a75:3611:7da2:9480:f8fd:31cc
Temporary IPv6 Address. . . . . : 2a02:7852:8a75:3611:e8c8:cfac:bcc1:7722
Link-local IPv6 Address . . . . . : fe80::7da2:9480:f8fd:31cc%7
IPv4 Address. . . . . : 10.0.2.10
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . : fe80::1:1%7
                           10.0.0.1
```

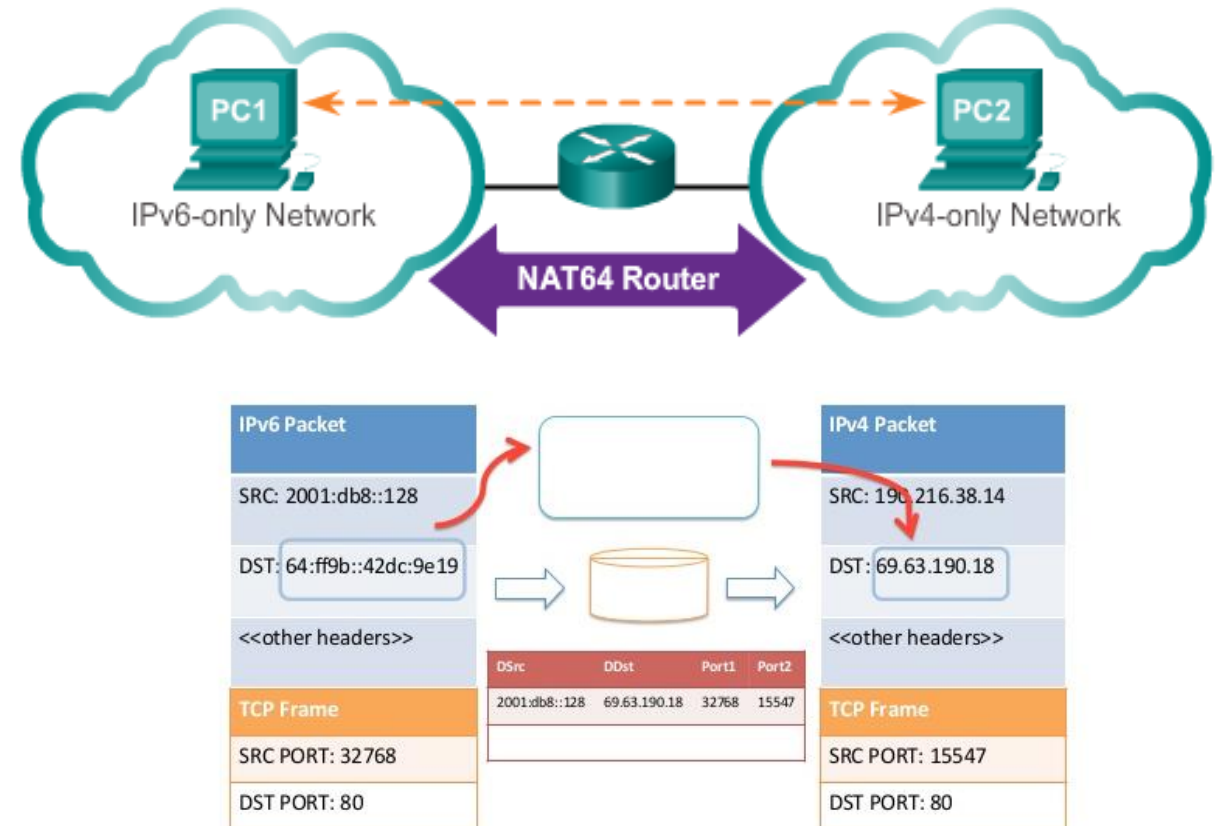
IPv4 and IPv6 Coexistence

Tunnelling: A method of transporting an IPv6 packet over an IPv4 network. The IPv6 packet is encapsulated inside an IPv4 packet.



IPv4 and IPv6 Coexistence

Translation: Network Address Translation 64 (NAT64) allows IPv6-enabled devices to communicate with IPv4-enabled devices using a translation technique similar to NAT for IPv4. An IPv6 packet is translated to an IPv4 packet, and vice versa.



sources

IPv6 Fundamentals: A Straightforward Aproach to Understanding IPv6

<https://www.ciscopress.com/articles/article.asp?p=2803866>

<https://slideplayer.com/slide/12713883/>